In [3]:
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# Detect outliers/anomalies in network traffic using density-based clustering (DBSC

from pathlib import Path

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import DBSCAN
from sklearn.decomposition import PCA
from sklearn.metrics import confusion_matrix, classification_report

# 1. Load the dataset
# The CSV file must be in the same folder as this Python file.
DATA_FILE = Path(r"D:\My Files\4th year\sem 7th\mlc lab\pract8\network_traffic_anom
df = pd.read_csv(DATA_FILE)

print("First 5 rows of dataset:")
print(df.head())

# 2. Select only numerical features for clustering
# DBSCAN is unsupervised, so we do not use the label column for training.
features = [
    "packet_size",
    "connection_duration",
    "src_bytes",
    "dst_bytes",
    "packet_rate",
    "error_rate"
]

X = df[features]

# 3. Standardize features
# Scaling is important because features such as src_bytes and error_rate
# have very different ranges.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 4. Apply DBSCAN clustering
# eps: radius around a point to search for neighbors
# min_samples: minimum number of nearby points required to form a dense region
model = DBSCAN(eps=1.0, min_samples=3)
df["cluster"] = model.fit_predict(X_scaled)

# In DBSCAN, cluster = -1 means noise/outlier/anomaly.
df["predicted_anomaly"] = df["cluster"].apply(lambda value: "anomaly" if value == -

# 5. Display clustering result
print("\nDBSCAN clustering result:")
print(df[["packet_size", "connection_duration", "src_bytes", "dst_bytes", "packet_r

# 6. Compare with actual labels if labels are available
```

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print("\nConfusion Matrix:")
print(confusion_matrix(df["label"], df["predicted_anomaly"], labels=["normal", "ano

print("\nClassification Report:")
print(classification_report(df["label"], df["predicted_anomaly"]))

# 7. Reduce dimensions using PCA for visualization
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

df["PCA1"] = X_pca[:, 0]
df["PCA2"] = X_pca[:, 1]

# 8. Plot normal points and anomalies
normal_points = df[df["predicted_anomaly"] == "normal"]
anomaly_points = df[df["predicted_anomaly"] == "anomaly"]

plt.figure(figsize=(8, 6))
plt.scatter(normal_points["PCA1"], normal_points["PCA2"], label="Normal")
plt.scatter(anomaly_points["PCA1"], anomaly_points["PCA2"], label="Anomaly", marker
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.title("Network Traffic Anomaly Detection using DBSCAN")
plt.legend()
plt.grid(True)
plt.show()

# 9. Save output result to a new CSV file
OUTPUT_FILE = Path(r"D:\My Files\4th year\sem 7th\mlc lab\pract8\dbscan_anomaly_out
df.to_csv(OUTPUT_FILE, index=False)
print(f"\nOutput saved as {OUTPUT_FILE.name}")
```

```
First 5 rows of dataset:
   packet_size  connection_duration  src_bytes  dst_bytes  packet_rate  \
0          450                   12       1200       3400           40
1          520                   15       1500       3600           42
2          480                   10       1100       3000           38
3          610                   18       1700       4200           45
4          560                   16       1600       3900           43

   error_rate   label
0        0.01  normal
1        0.02  normal
2        0.01  normal
3        0.03  normal
4        0.02  normal

DBSCAN clustering result:
    packet_size  connection_duration  src_bytes  dst_bytes  packet_rate  \
0           450                   12       1200       3400           40
1           520                   15       1500       3600           42
2           480                   10       1100       3000           38
3           610                   18       1700       4200           45
4           560                   16       1600       3900           43
5           430                   11       1000       3100           37
6           590                   17       1650       4000           44
7           500                   14       1400       3500           41
8           470                   13       1300       3300           39
9           620                   20       1800       4300           46
10          300                   50        500      20000            8
11          320                   55        600      21000            9
12          310                   48        550      19000            8
13          290                   52        520      20500            7
14          305                   49        580      19800            8
15          250                   80        400      30000            4
16          260                   75        450      28000            5
17          255                   78        420      29000            4
18          270                   76        470      28500            5
19          240                   82        390      31000            4
20          900                    3       9000        400          250
21         1200                    1      15000        100          350
22         1700                    8      26000       8000           90
23         2000                    1      50000         10          800
24          700                  100        200      50000            2
25           50                    5        100         50          600

    error_rate  cluster predicted_anomaly
0         0.01        0            normal
1         0.02        0            normal
2         0.01        0            normal
3         0.03        0            normal
4         0.02        0            normal
5         0.01        0            normal
6         0.02        0            normal
7         0.02        0            normal
8         0.01        0            normal
9         0.03        0            normal
10        0.04        1            normal
```

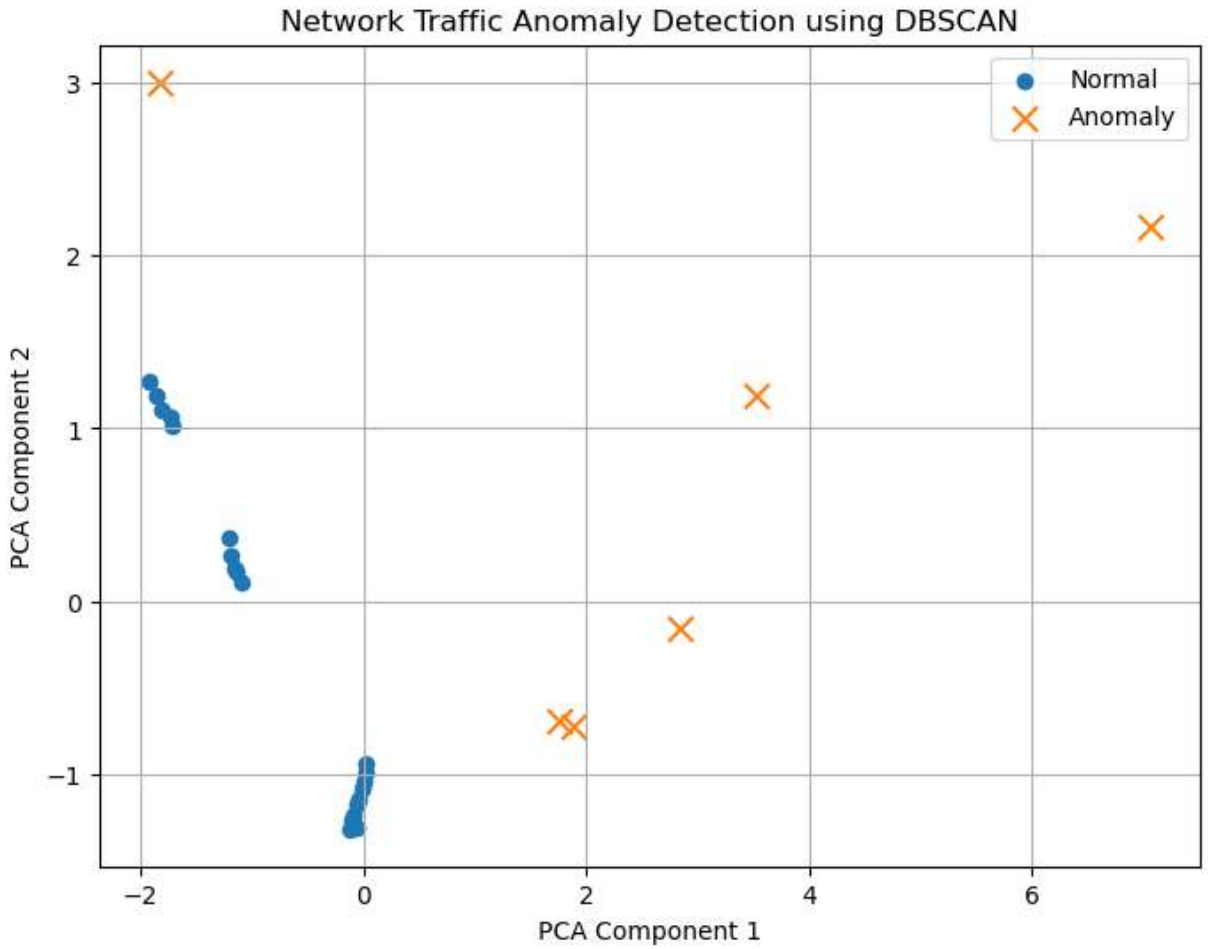
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11        0.05        1            normal
12        0.04        1            normal
13        0.05        1            normal
14        0.04        1            normal
15        0.02        1            normal
16        0.03        1            normal
17        0.02        1            normal
18        0.03        1            normal
19        0.02        1            normal
20        0.25       -1           anomaly
21        0.40       -1           anomaly
22        0.70       -1           anomaly
23        0.90       -1           anomaly
24        0.20       -1           anomaly
25        0.55       -1           anomaly

Confusion Matrix:
[[20  0]
 [ 0  6]]

Classification Report:
              precision    recall  f1-score   support

     anomaly       1.00      1.00      1.00         6
      normal       1.00      1.00      1.00        20

    accuracy                           1.00        26
   macro avg       1.00      1.00      1.00        26
weighted avg       1.00      1.00      1.00        26
```




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Output saved as dbscan_anomaly_output.csv
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In [ ]: